



SACS

Interactive Fatigue

Version 24.00

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1 INTRODUCTION

1.1 GENERAL INFORMATION

The Interactive Fatigue program allows for reviewing of tubular connection fatigue results and redesigning connections in a 3D interactive environment. The program requires a fatigue extract file generated by the Fatigue batch program module as input. Various reports and plots of fatigue results can be generated and saved.

1.1.1 Program Features

Some of the main features of the Interactive Fatigue program are:

1. Shows the 3-D view of the connection and allows for braces to be selected with the mouse.
2. Allows for redesigning of chord and brace members based on fatigue results.
3. Allows SCF theory to be changed for any type connection, including in-line connections and connections with user defined SCFs.
4. Provides the ability to designate the S-N curve used to determine allowable cycles.
5. Recognizes all SCF and S-N options available in the batch program.
6. Full reporting capabilities.
7. Full plotting capabilities, including transfer function, response function, wave spectrum, stress range versus cycles and various other plots.
8. Reports and plots can be displayed on the screen or saved to a file.

1.2 PROGRAM OVERVIEW

1.2.1 File Naming Conventions

While no file naming convention is required for Interactive Fatigue, the file name filter is used in the Open dialog box depending on the naming convention set in SACS Executive. The default filter is *???ext.** for the naming convention *pgmtyp.ver* and **.ext* for *verpgm.typ*. Plot filenames may be specified by the user using Save NPF As under the File menu, and reports may be saved from the SACS report viewer program which is automatically called when reports are requested from the Reports menu.

2 GETTING STARTED

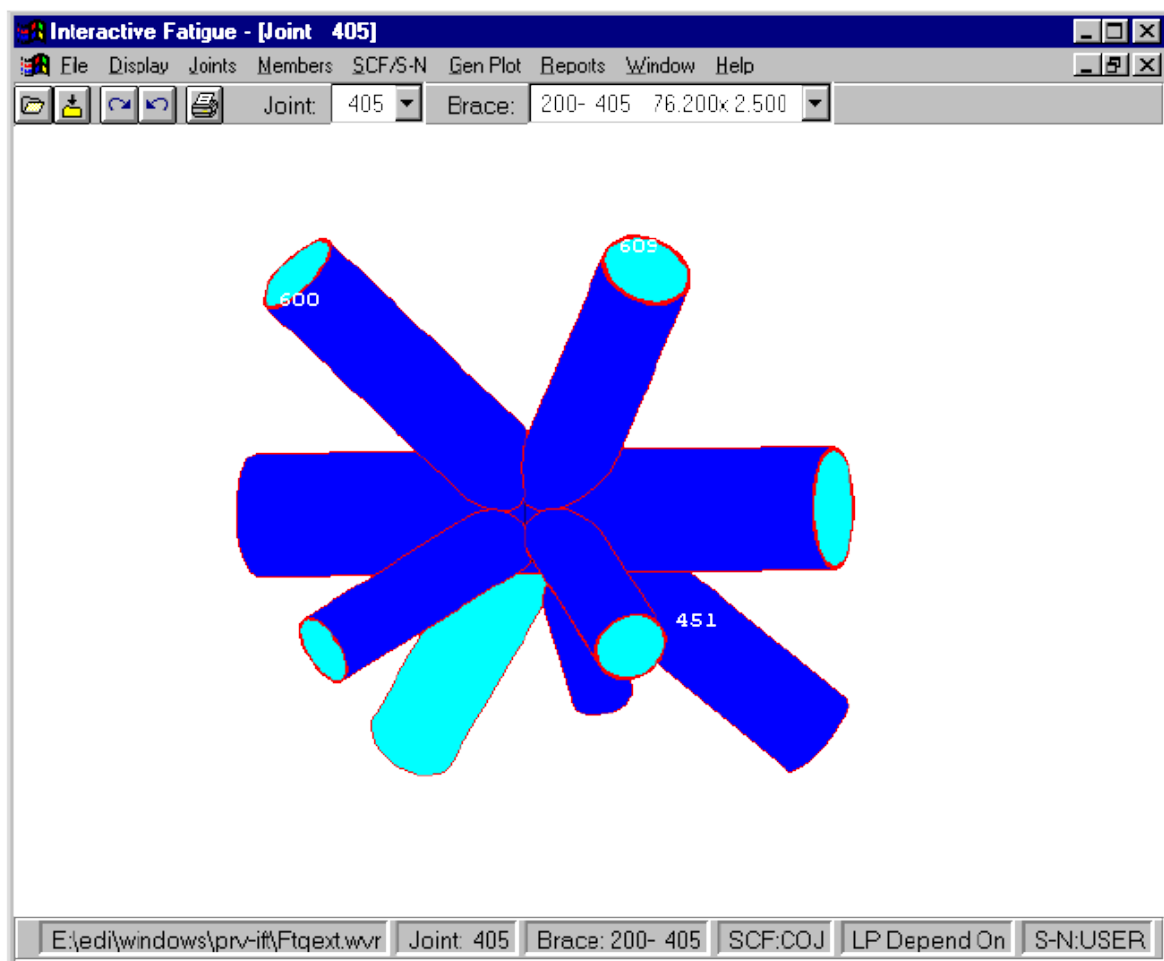
2.1 BEGINNING AN INTERACTIVE FATIGUE SESSION

To start the Interactive Fatigue program, type "ifatigue *filename*" where *filename* is the Fatigue extract file name. Interactive Fatigue can also be launched from the SACS Executive by selecting Interactive Fatigue from the menu then selecting an extract file name.

If an extract file name is not selected, the program is initiated with a blank screen. An extract file can then be selected using the File/Open menu selection.

When a Fatigue extract file is opened, Interactive Fatigue initially displays the first joint of the file with the first brace listed shown as the selected brace. Different joints may be selected using the Joints/Select menu option. Braces may be selected using the Members/Select Brace menu option or by simply clicking on the desired brace with the mouse.

Various display features, such as fill, double line drawing, labeling, and rotate are available in the Display Menu.



2.2 FATIGUE CONNECTION COORDINATE SYSTEM

For each SACS load case making up the fatigue environment, fatigue results are evaluated at eight points spaced at 45 degrees around the perimeter of the welded connection on both the chord and brace side, see Figure 1 below:

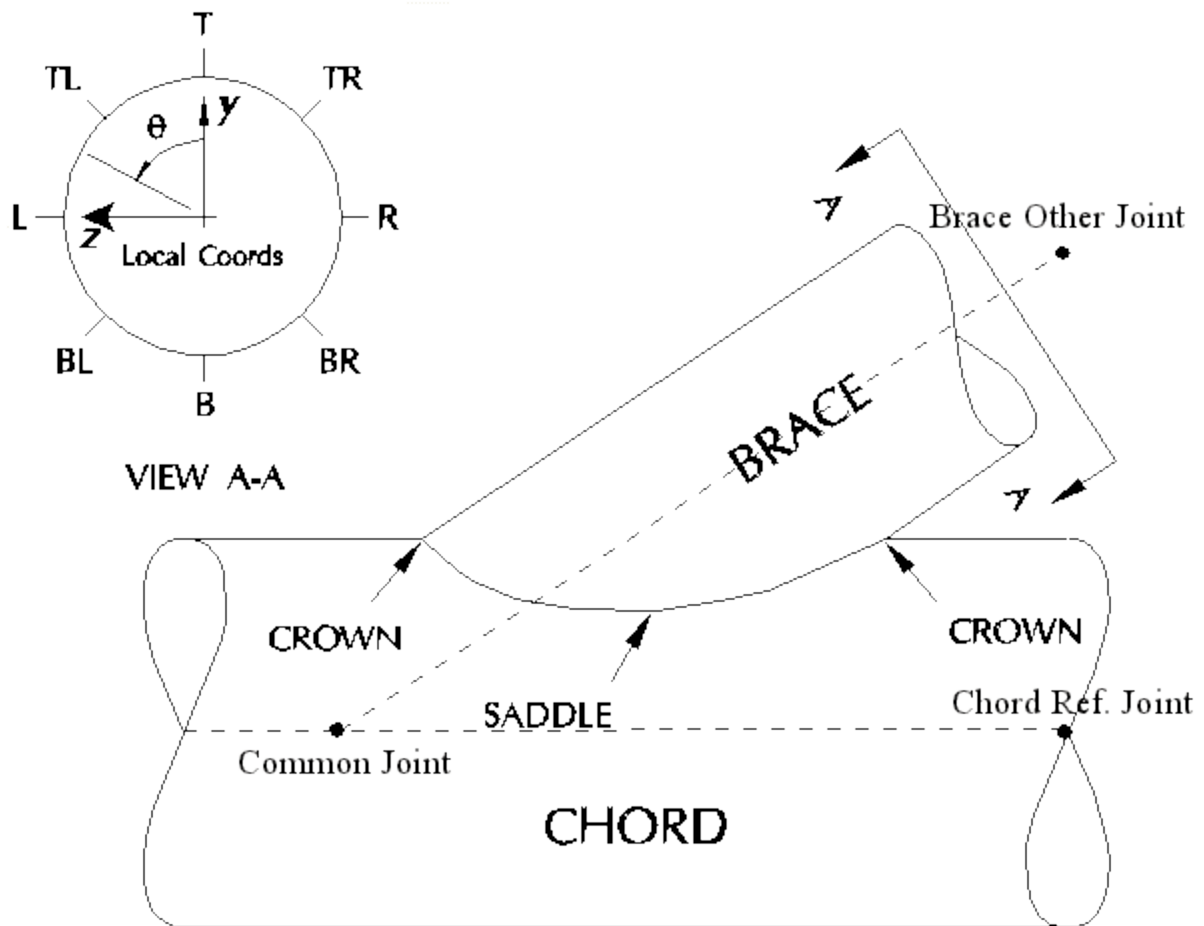


Figure 1

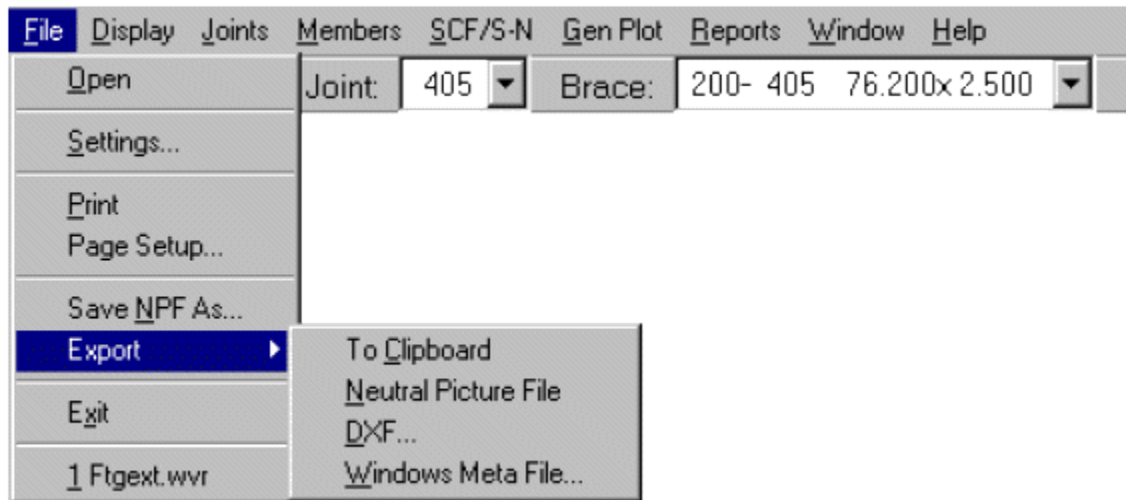
Note: The top of the connection is at the intersection of the brace/chord plane opposite of the chord reference joint. The Bottom position is therefore always at the intersection of the brace/chord plane on the side nearest to the chord reference joint.

2.3 EXISTING THE INTERACTIVE FATIGUE SESSION

Select the File/Exit menu option to exit the Interactive Fatigue session. If any plots or reports that have been generated to a file have not been saved, the user is prompted whether or not to save the file.

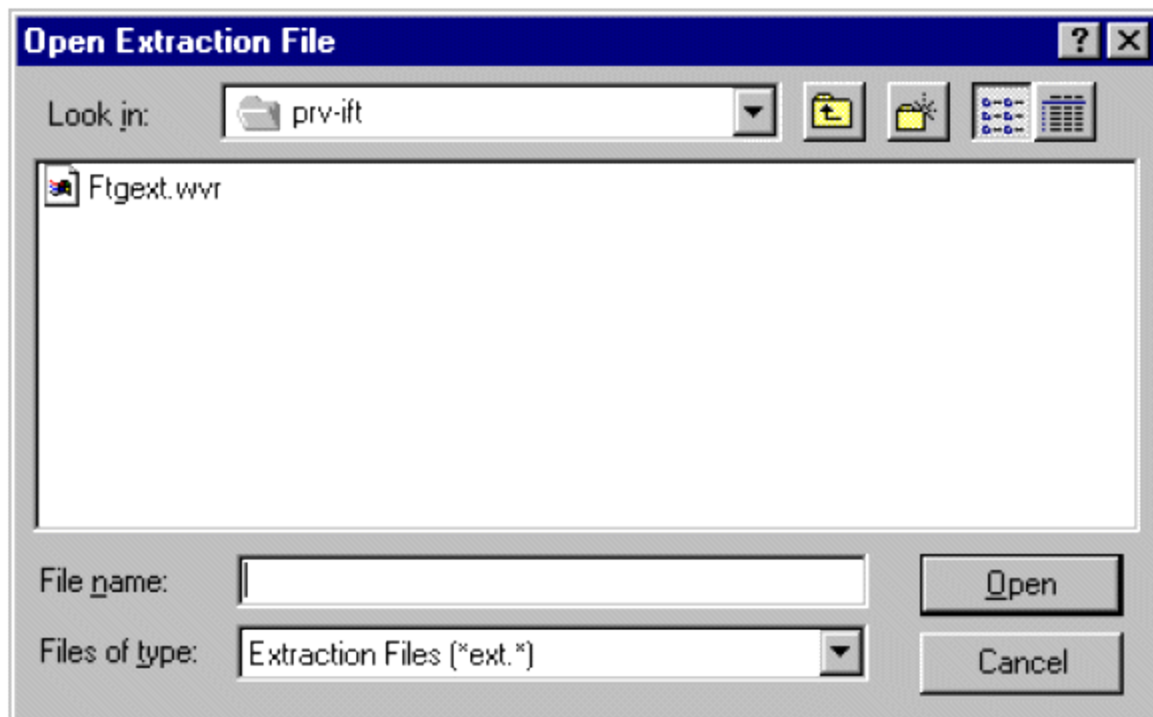
3 FILE MENU FEATURES

The following section details the commands found under the File main menu.



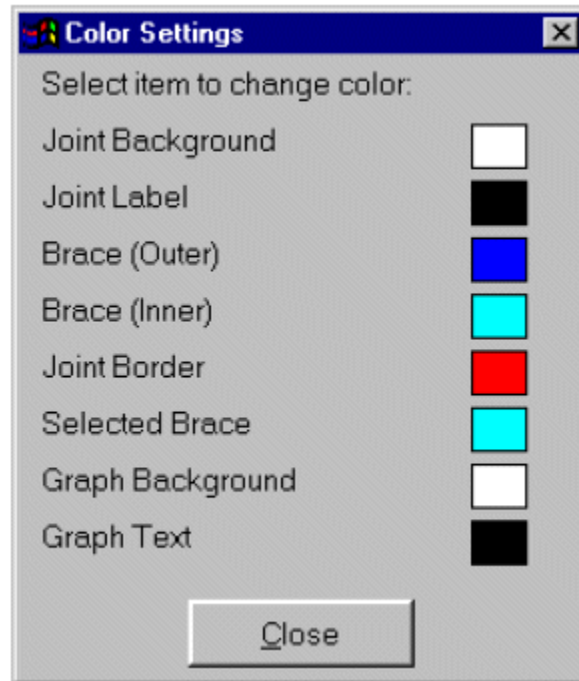
3.1 OPEN

Open is used to import an existing Fatigue extract file into Interactive Fatigue.



3.2 SETTINGS

Settings allows the display colors to be specified for the background and text of the joint images and the graphs, and also the coloring of the inner and outer surfaces of the braces, the selected brace color, and the border color around the joint outline.



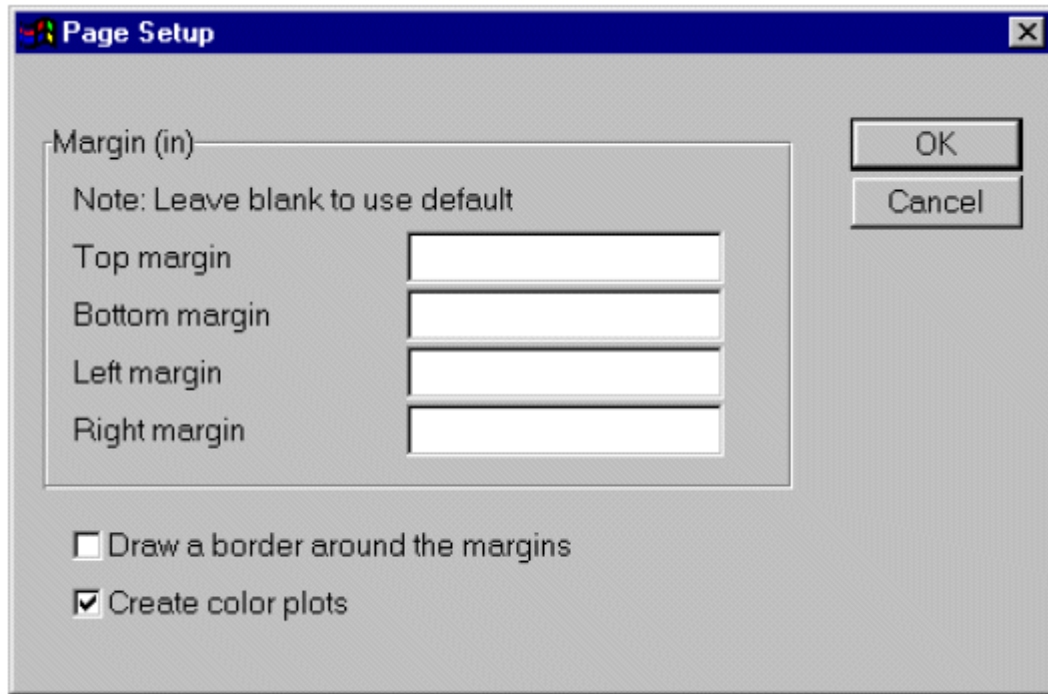
Note: Modifications to the program settings affect only the current Interactive Fatigue session and are not saved to the configuration file.

3.3 PRINT

The Print option is used to print the current plot display.

3.4 PAGE SETUP

The Page Setup dialog provides for margin, border and color plot settings.



3.5 SAVE NPF

The Save NPF option is used to save plots that are generated to a neutral picture file.

3.6 EXPORT

The Export option allows the choice of exporting the plot information in a variety of formats.

3.6.1 To Clipboard

This copies the current plot to the Windows clipboard to allow pasting into other Windows applications.

3.6.2 Neutral Picture File

The Neutral Picture File format is a SACS specific plot format that can be read by the Prevue program.

3.6.3 DXF

The Data eXchange Format (DXF) format is widely supported by CAD and other graphics programs.

3.6.4 Windows Meta File

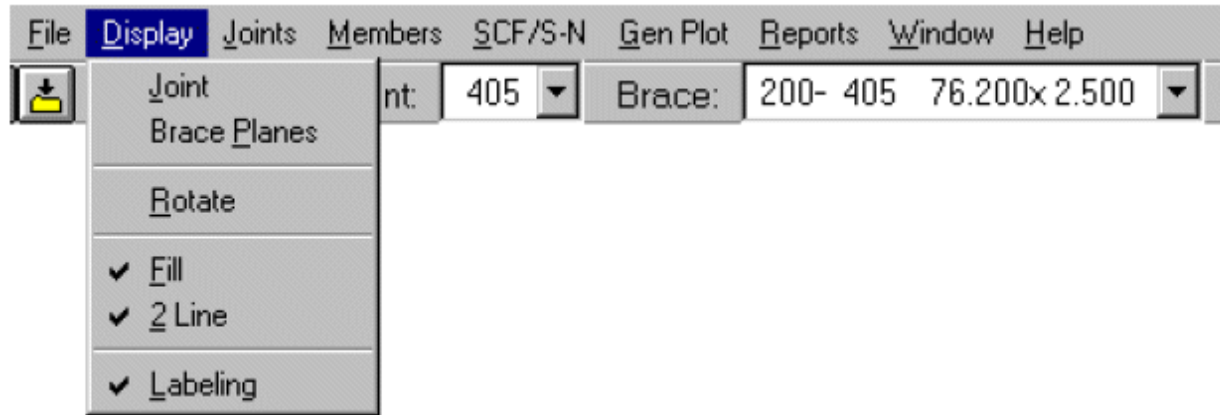
The Windows Meta File (WMF) format is supported by many Windows graphics and word-processing programs.

3.7 EXIT

The Exit command terminates the Interactive Fatigue session.

4 DISPLAY MENU FEATURES

The Display Menu features allow the user to specify or select display options including joint 3D view, joint plane view, rotation, fill, and labeling.



4.1 JOINT

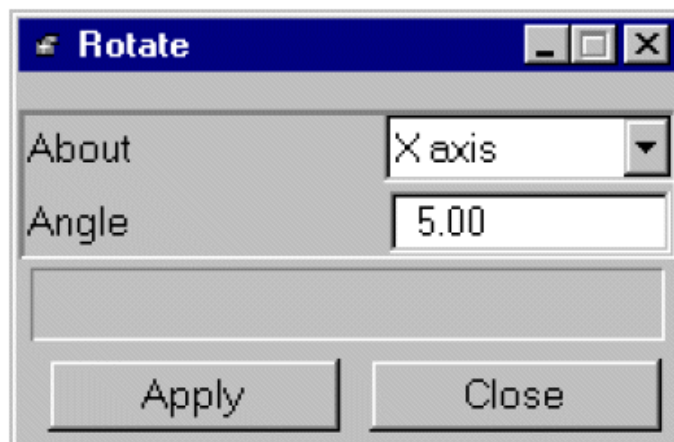
The Joint command displays the 3D view of the currently selected joint.

4.2 BRACE PLANES

The Brace Planes selection shows the various joint planes formed by the chord and braces of the selected joint.

4.3 ROTATE

Rotate allows rotation of the 3D joint view about the major axes by the specified angle in degrees.



4.4 FILL

Fill toggles polygon filling in the 3D joint and brace plane views.

4.5 '2 LINE'

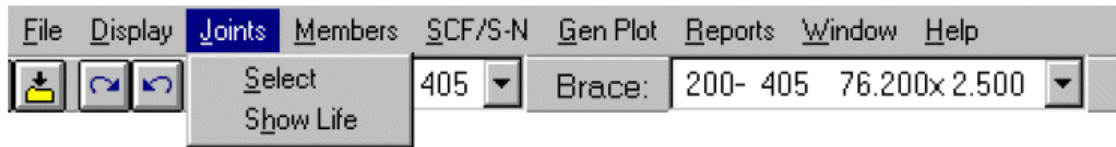
The 2 Line option toggles between single line and double line display of the joint.

4.6 LABELING

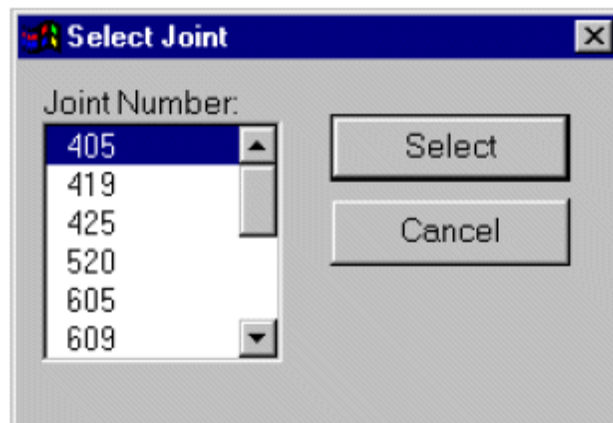
Selecting the Labeling command toggles the joint label of braces on and off. The joint label represents the joint name of the end of the brace not connected to the selected joint.

5 JOINTS MENU FEATURES

The Joints Menu features allow the user to designate the current joint and show fatigue life results.



5.1 SELECT

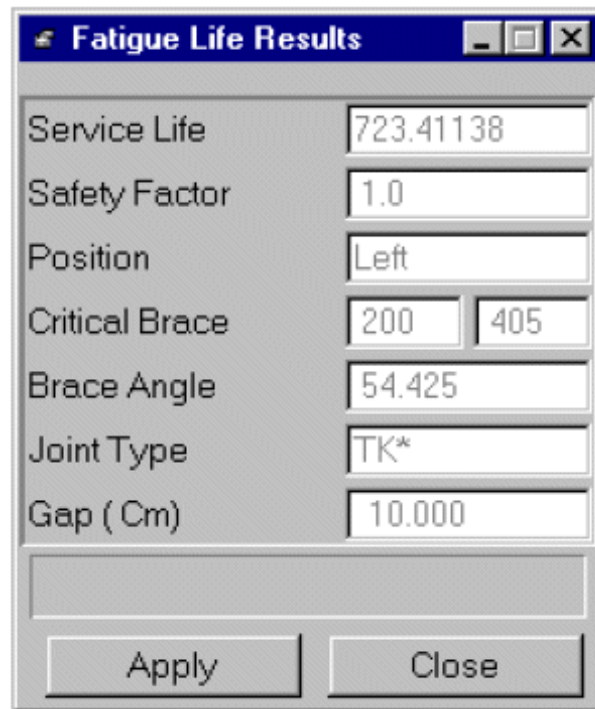


Select brings up a list of the available joints in the extract file from which to select the current joint.

Upon selecting a joint, the first brace in the brace list is selected as the active brace. The SCF, S-N and load dependency settings are set to the original fatigue batch analysis settings specified on the FTOPT line.

5.2 SHOW LIFE

The Show Life command displays the fatigue life or service life results for the currently selected joint and brace based on the current SCF, S-N and load path dependency settings.

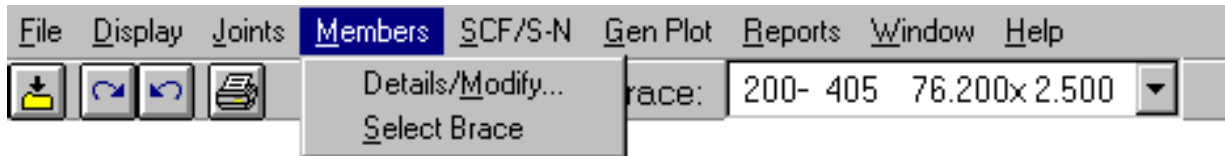


The image shows a software dialog box titled "Fatigue Life Results". It contains several input fields for fatigue analysis parameters. The fields are: "Service Life" with a value of 723.41138, "Safety Factor" with a value of 1.0, "Position" with a value of "Left", "Critical Brace" with two sub-fields containing 200 and 405, "Brace Angle" with a value of 54.425, "Joint Type" with a value of "TK*", and "Gap (Cm)" with a value of 10.000. At the bottom of the dialog are two buttons: "Apply" and "Close".

Parameter	Value
Service Life	723.41138
Safety Factor	1.0
Position	Left
Critical Brace	200 405
Brace Angle	54.425
Joint Type	TK*
Gap (Cm)	10.000

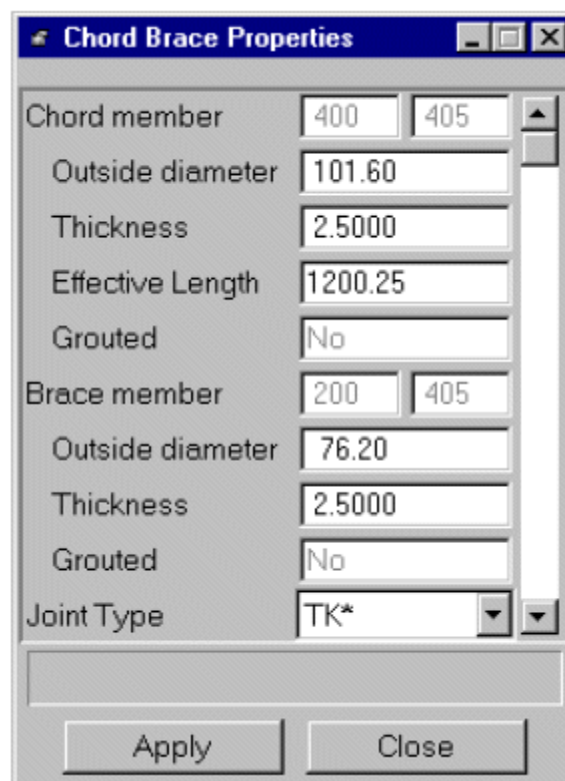
6 MEMBERS MENU FEATURES

The Members Menu features allow the user to designate the current or active brace and to view or modify chord and brace properties.



6.1 DETAILS/MODIFY

The Details/Modify command displays property information for the chord and the current or active brace member.

A screenshot of a dialog box titled 'Chord Brace Properties'. It contains two sections: 'Chord member' and 'Brace member'. Each section has fields for 'Outside diameter', 'Thickness', 'Effective Length', and 'Grouted'. The 'Joint Type' is set to 'TK*'. At the bottom are 'Apply' and 'Close' buttons.

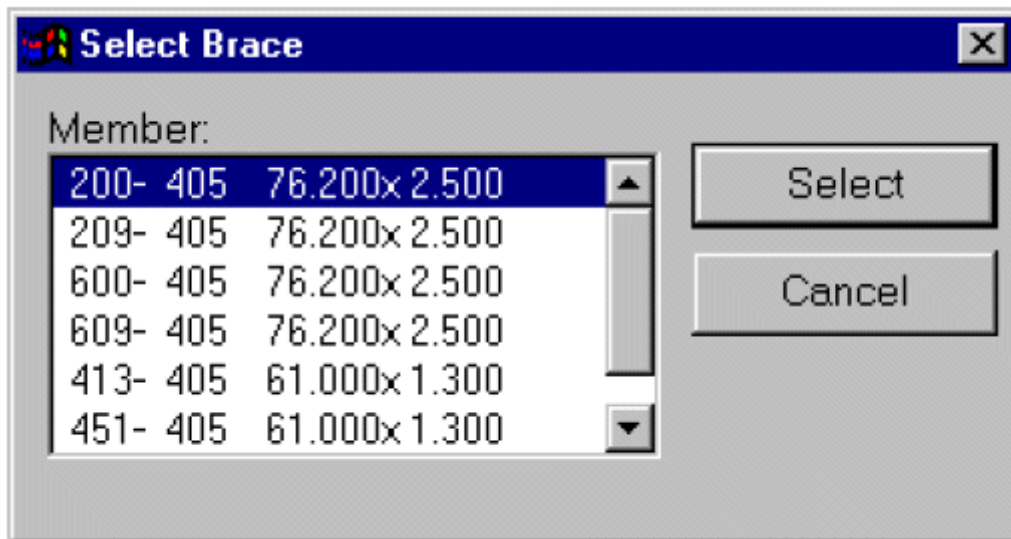
Chord member	
Outside diameter	101.60
Thickness	2.5000
Effective Length	1200.25
Grouted	No

Brace member	
Outside diameter	76.20
Thickness	2.5000
Grouted	No

Joint Type: TK*

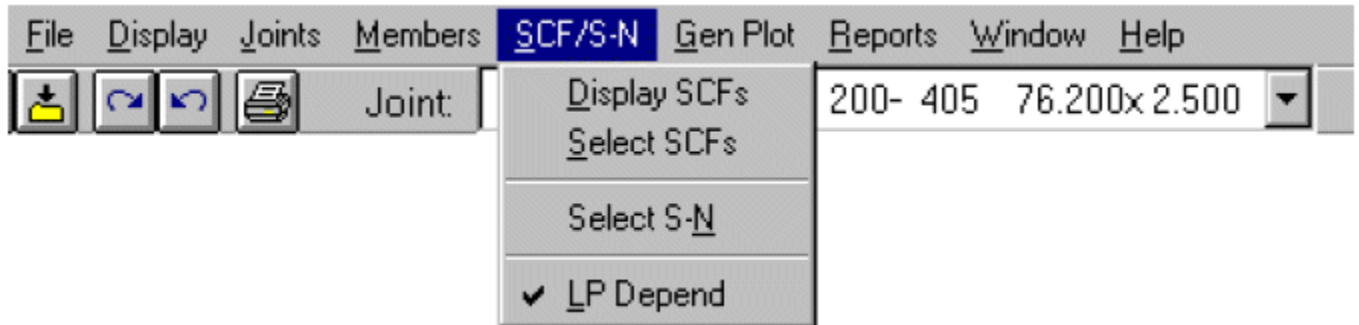
6.2 SELECT BRACE

The Select Brace option displays a list of braces connected to the active joint from which the user may designate as the active brace member.



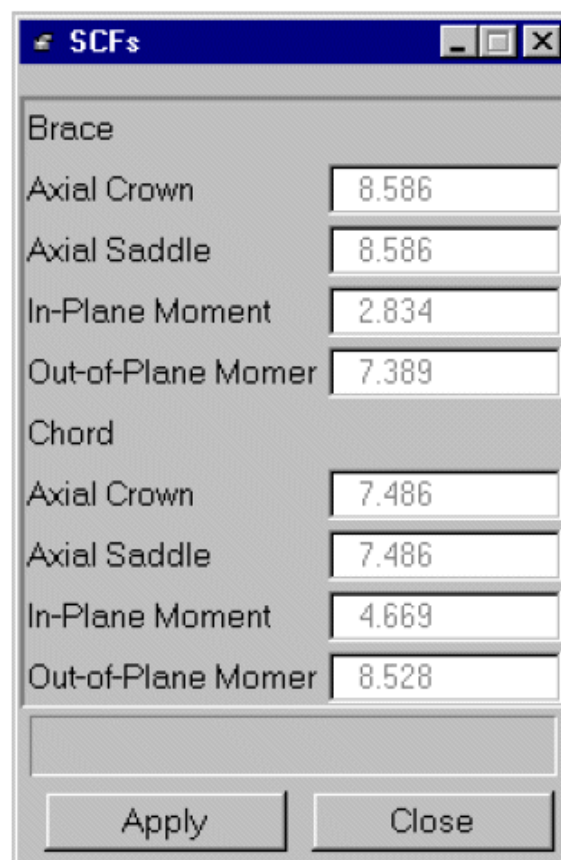
7 SCF/S-N MENU FEATURES

The SCF/S-N menu features allow the user to display and/or modify the current Stress Concentration Factors theory, S-N curve and load path dependency settings.



7.1 DISPLAY SCFS

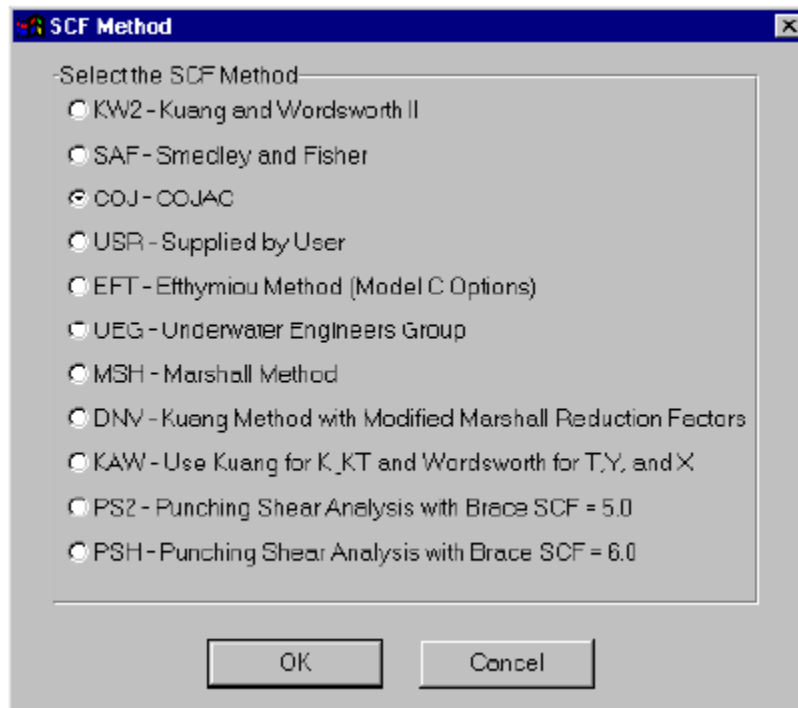
The Display SCFs command displays the Stress Concentration Factors for the chord and the active brace.



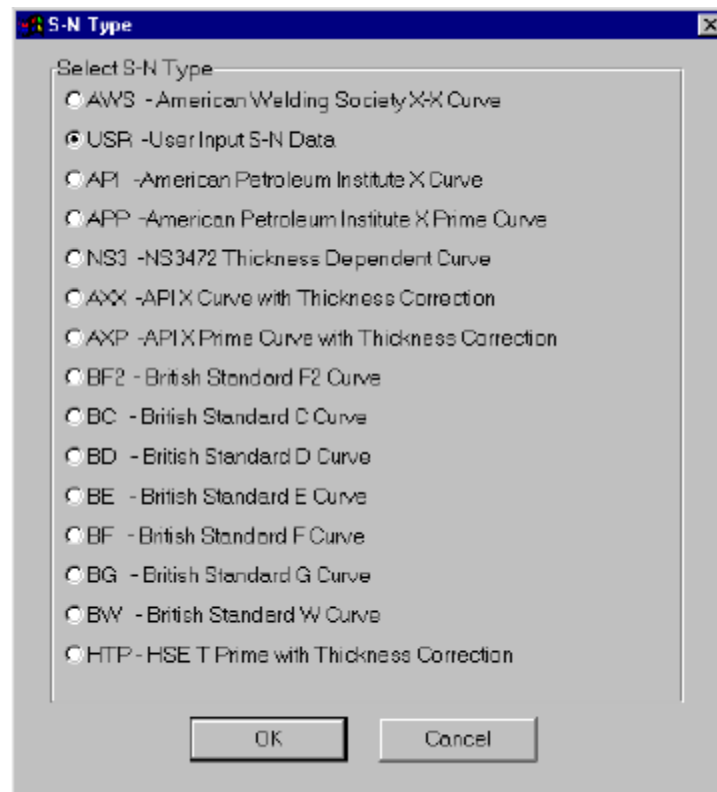
7.2 SELECT SCFS

The Select SCFs option allows designation of the current SCF theory for which to calculate chord and brace SCFs.

Select the desired SCF theory from the list of available SCF methods.



7.3 SELECT S-N



The Select S-N command allows for the designation of the S-N curve to be used when determining the allowable number of stress range cycles.

To specify the current S-N curve, select the radio button next to the S-N option desired.

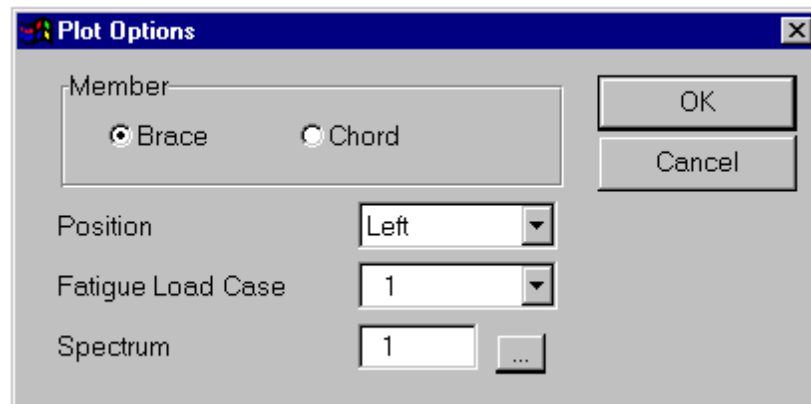
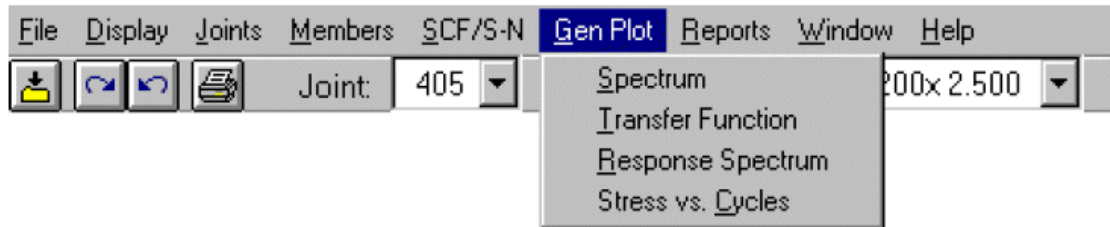
7.4 LP DEPEND

The LP Depend command is a toggle that designates whether or not SCFs are to be calculated based on load path.

Note: The current load path dependance setting is displayed on the status bar.

8 GEN PLOT MENU FEATURES

The Gen Plot menu features allow the user to generate various plots and graphs for display on the screen or to save in a plot file.



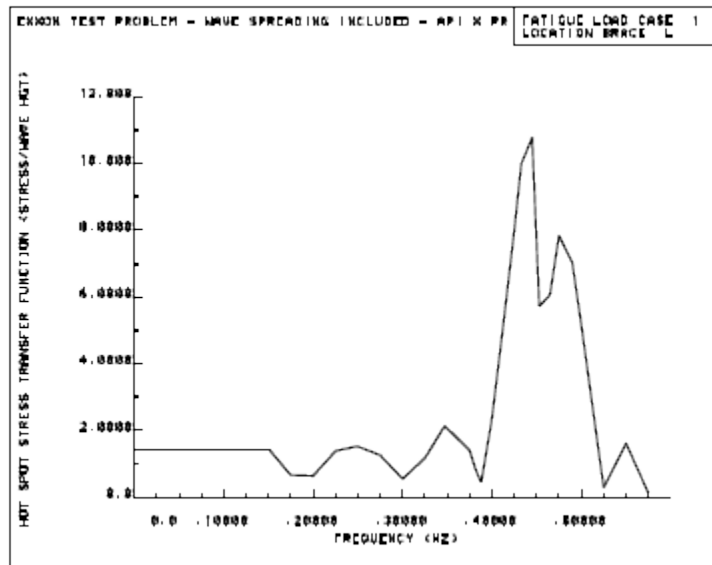
Upon selection of any of the plot commands, the Plot Options dialog is displayed. Select the desired plot options including the element to be plotted, i.e. chord or brace, and position around the circumference. The desired Fatigue load case and if applicable the desired wave spectrum may be designated.

Note: Plots are generated and displayed on the screen by default. To save the currently displayed plot to a file, select the File/Plot command.

8.1 SPECTRUM

The Spectrum plot command plots the wave spectrum designated on the Plot Options dialog. The Member and Position plot options are not applicable to this feature.

8.2 TRANSFER FUNCTION



Select the Transfer Function command to plot the cyclic hot-spot stress transfer function of the chord or brace position designated and for the particular Fatigue Load Case (wave direction) specified on the Plot Options dialog.

8.3 RESPONSE SPECTRUM

Select the Response Spectrum command to plot the response spectrum of the chord or brace position designated. The applicable Fatigue Load Case and Wave spectrum are designated on the Plot Options dialog.

8.4 STRESS VS. CYCLES

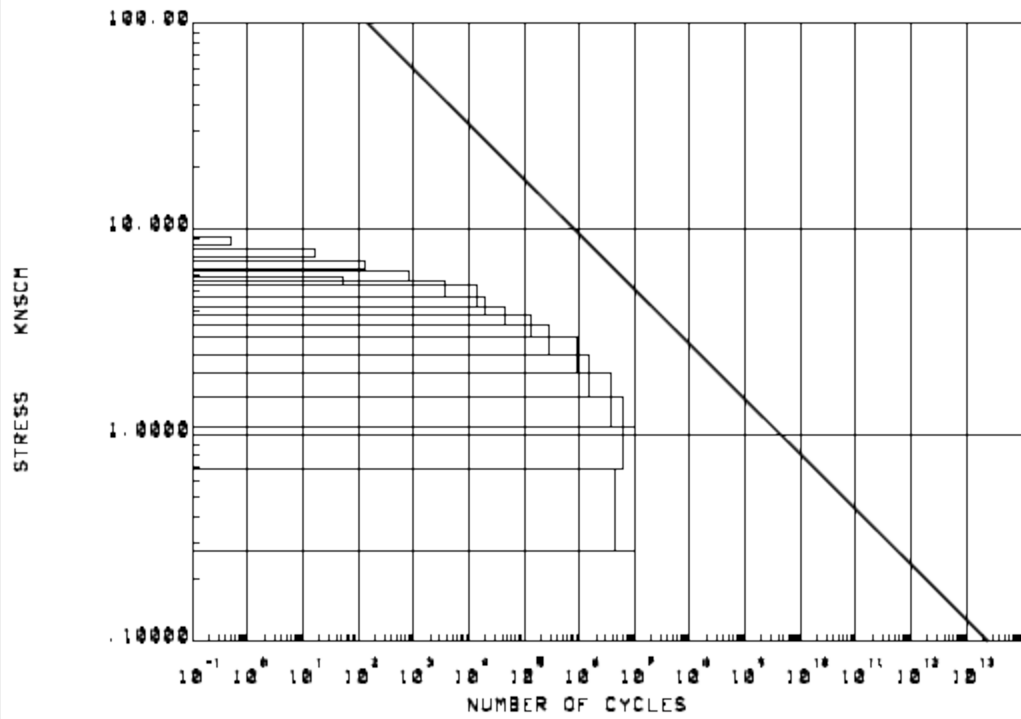
The Stress vs. Cycles command creates a bar graph showing stress range and number of cycles plotted against the S-N allowable curve. The chord or brace position along with the applicable fatigue load case must be designated on the Plot Options dialog.

The Stress vs. Cycles plot is generated as follows:

For a point on the connection, the maximum cyclic stress is found considering all spectra for the direction in question. This maximum stress is then broken into 20 cyclic stress bands or categories. For each band, the total damage and number of cycles is summed. The report and plot shows the range and cycles for a band only if the band causes actual damage.

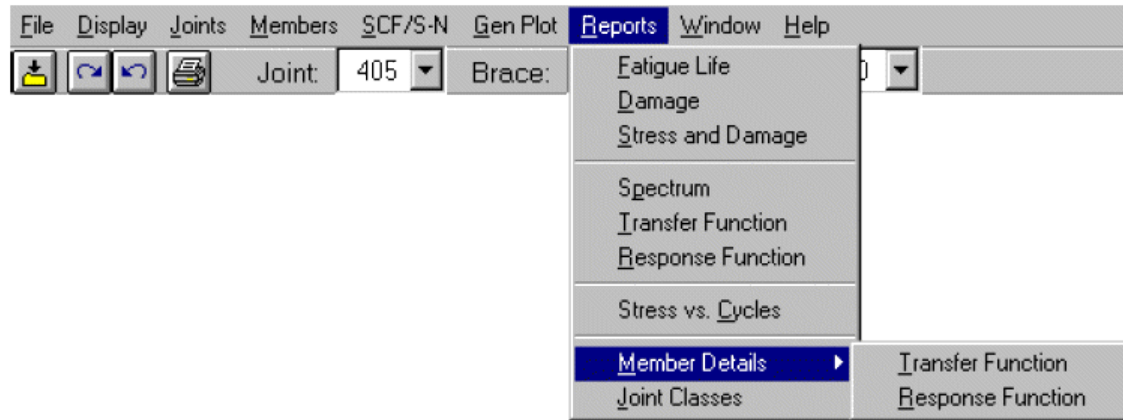
EXXON TEST PROBLEM - WAVE SPREADING INCLUDED - API X PR

CHORD 400- 405
BRACE 200- 405
POINT BRACE L
FATIGUE LOAD CASE 1



9 REPORTS MENU FEATURES

The Reports Menu features allow the user to generate various reports.



Reports may be displayed on the screen in the Report Browser and from there may be saved to a file.

9.1 FATIGUE LIFE

The Fatigue Life command creates a detailed report for the active joint and brace. The report contains basic properties of the chord and brace, fatigue life and the damage reported at each of the eight positions. The following is a sample Fatigue Life report:

```

Fatigue Life results at Joint 101
Brace 101- 203 Chord 101- 201

Fatigue Life (years)      796.385      SCF Method      Efthymiou Method
Design Life (years)      20.000      Life Safety Fact  2.000
Maximum Damage           0.025      Location         Bottom
Brace-Chord Angle (deg)  30.913      Connection Type   K      Gap ( Cm)  14.003

```

*** Brace and Chord Properties ***

Member	Chord Brace	Diameter (Cm)	Thick (Cm)	*** Stress Concentration Factors ***			
				Ax-Crn	Ax-Sdl	IP-Mom	OP-Mom
101- 201	Chord	105.000	3.500	4.204	2.919	1.151	1.495
101- 203	Brace	50.000	2.000	3.933	2.265	2.800	1.399

*** Damage and Fatigue Life Details ***

Position	***** CHORD *****		***** BRACE *****	
	Damage	Life	Damage	Life
Top	.023662	845.236	.013024	1535.67
Top Left	.7728-2	2588.11	.2266-2	8827.50
Left	.5764-3	34700.5	.1230-4	1626.+3
Bottom Left	.7998-2	2500.55	.3062-2	6530.88
Bottom	.025113	796.385	.019896	1005.23
Bottom Right	.7791-2	2567.04	.2965-2	6745.00
Right	.5730-3	34902.6	.1328-4	1506.+3
Top Right	.7542-2	2651.88	.2652-2	7540.43
Maximum	.025113	796.385	.019896	1005.23

9.2 DAMAGE

The Damage report command generates a detailed report showing damage at each of the eight positions around the circumference of the connection caused by each wave spectrum.

```

Fatigue Damage Details at Joint 101
Chord 101- 201 Brace 101- 203

Ftg  Spect  Chord  Top      Top      Left      Bottom   Bottom   Bottom   Right      Top
LC   No.   Brace  Top      Left      Left      Left      Left      Right      Right      Right
-----
1     1     C .0000000 .0000000 .0000000 .0000000 .0000000 .0000000 .0000000 .0000000
      B .0000000 .0000000 .0000000 .0000000 .0000000 .0000000 .0000000 .0000000

1     2     C .19072-4 .14068-5 .0000000 .11842-5 .15504-4 .12271-5 .0000000 .14561-5
      B .17551-4 .0000000 .0000000 .0000000 .14194-5 .0000000 .0000000 .0000000

2     1     C .0000000 .0000000 .0000000 .0000000 .0000000 .0000000 .0000000 .0000000
      B .0000000 .0000000 .0000000 .0000000 .0000000 .0000000 .0000000 .0000000

```

9.3 STRESS AND DAMAGE

The Stress and Damage reports shows RMS stress and Damage at each chord and brace position around the circumference of the connection for each of the wave spectra in the Fatigue Load case specified by the user.

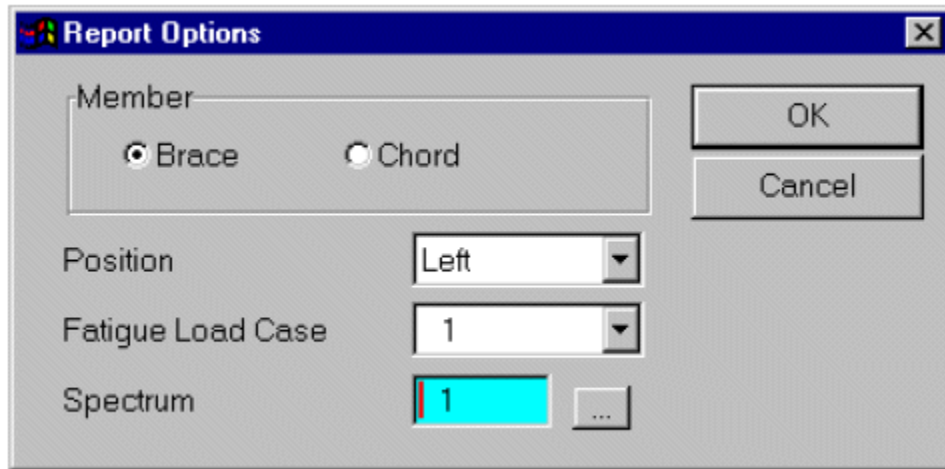
Fatigue RMS Stress and Damage at Joint 101					
		Chord 101- 201		Brace 101- 203	
Fatigue Load Case	Spectrum Number	Position	Chord Brace	RMS Stress (KG/CM2)	Damage
1	1	Bottom	Brace	0.234624	.14194-5
	1	Bottom Left	Brace	0.190174	.0000000
	1	Left	Brace	0.138432	.0000000
	1	Top Left	Brace	0.219104	.0000000
	1	Top	Brace	0.275573	.17551-4
	1	Top Right	Brace	0.221414	.0000000
	1	Right	Brace	0.141655	.0000000
	1	Bottom Right	Brace	0.192416	.0000000
	1	Bottom	Chord	0.263619	.15504-4
	1	Bottom Left	Chord	0.224893	.11842-5
	1	Left	Chord	0.181843	.0000000
	1	Top Left	Chord	0.236798	.14068-5
	1	Top	Chord	0.280472	.19072-4
	1	Top Right	Chord	0.239245	.14561-5
	1	Right	Chord	0.185285	.0000000
	1	Bottom Right	Chord	0.227313	.12271-5
1	2	Bottom	Brace	0.291327	.16509-2
	2	Bottom Left	Brace	0.237372	.32396-4
	2	Left	Brace	0.163807	.0000000
	2	Top Left	Brace	0.237601	.33213-4
	2	Top	Brace	0.291674	.16994-2
	2	Top Right	Brace	0.233524	.31164-4
	2	Right	Brace	0.158029	.0000000
	2	Bottom Right	Brace	0.233282	.30377-4
	2	Bottom	Chord	0.311029	.21247-2
	2	Bottom Left	Chord	0.267762	.36991-3
	2	Left	Chord	0.213636	.0000000
	2	Top Left	Chord	0.267857	.37329-3
	2	Top	Chord	0.311171	.21485-2
	2	Top Right	Chord	0.263496	.35137-3
	2	Right	Chord	0.207463	.0000000
	2	Bottom Right	Chord	0.263396	.34811-3

9.4 SPECTRUM

The Spectrum report shows the points of the wave spectrum selected by the user in tabular format.

9.5 TRANSFER FUNCTION

The Transfer Function report shows the points of the transfer function for the selected point on the chord or brace. The chord or brace position along with the Fatigue load case are specified on the Report Options dialog.



The 'Report Options' dialog box contains the following fields and controls:

- Member:** Radio buttons for 'Brace' (selected) and 'Chord'.
- Position:** A dropdown menu currently showing 'Left'.
- Fatigue Load Case:** A dropdown menu currently showing '1'.
- Spectrum:** A text field showing '1' and a button with three dots '...'.
- Buttons:** 'OK' and 'Cancel' buttons on the right side.

Note: Designation of the wave spectrum is not required for the transfer function report.

Transfer Function Report				
Fatigue Load Case 1 Chord 101- 201 Position Bottom				
Wave	Height (M)	H.S. Stress Range (KGSCM)	HSSR/Height (KGSCM *M)	Frequency (Hz)
1	10.300	148.953	14.461	0.10000
2	6.620	111.227	16.802	0.12500
3	3.720	77.895	20.940	0.16667
4	2.590	34.808	13.440	0.20000
5	1.650	56.084	33.991	0.25000
6	0.930	103.111	110.872	0.33333
7	0.520	130.998	251.920	0.44444
8	0.410	57.396	139.992	0.50000
9	0.320	144.331	451.035	0.57143
10	0.230	54.496	236.941	0.66667

9.6 RESPONSE FUNCTION

The Response Function report shows the points of the response function for the selected point on the chord or brace for the specified wave spectrum. The chord or brace position along with the Fatigue load case and the appropriate wave spectrum are specified on the Report Options dialog.

9.7 STRESS VS. CYCLES

Stress Range vs. Cycle Report			
Fatigue Load Case 1 Chord 101- 201 Position Bottom			
	Range Number	Stress Range (KGSCM)	Cycles
	-----	-----	-----
	9	257.1	
1515367			
	10	288.7	481573
	11	314.7	231189
	12	347.3	143603
	13	354.6	17075
	14	396.2	39501
	15	423.9	6830
	16	453.8	3096
	17	486.9	1536
	19	552.7	153

The Stress vs. Cycles report contains hot-spot stress ranges and number of cycles for the selected point on the chord or brace. The chord or brace position along with the Fatigue load case are specified on the Report Options dialog.

9.8 MEMBER DETAILS

Member detail reports contain detailed information for the chord or brace member specified.

9.8.1 Transfer Function

The transfer function detail report contains axial load and bending moment normalized by wave height. The report also contains real and imaginary roots.

Transfer Function Detail Report									
Member 101- 203 Fatigue Load Case 1 Member End 101									
Wave	Wave	AXIAL/	OP-MOM/	IP-MOM/	** AXIAL LOAD **		OUT-PLANE MOMENT		
IN-PLANE	MOMENT				REAL	IMAG	REAL	IMAG	REAL
Height	Freq	HEIGHT	HEIGHT	HEIGHT	REAL	IMAG	REAL	IMAG	REAL
IMAG									
M	Hz	KG/M	CM-KG/M	CM-KG/M	KG/M	CM-KG/M	CM-KG/M	KG/M	CM-KG/M
CM-KG/M									
-----	-----	-----	-----	-----	-----	-----	-----	-----	-----
10.300	0.1000	10493.836	4479.517	27358.572	4930.489	1794.555	-2104.684	766.043	13679.286
0.000									
6.620	0.1250	7616.267	3063.282	6784.224	2917.198	2447.823	-1439.272	523.853	2598.509
2180.407									
3.720	0.1667	5160.029	1618.978	4113.219	1290.006	2234.359	-809.489	0.000	-357.126
-2025.365									
2.590	0.2000	2264.610	692.326	5409.069	-196.623	1115.103	-265.176	-222.509	-2071.793
-1738.442									
1.650	0.2500	3707.031	1250.661	6416.890	321.860	-1825.356	-587.618	-213.876	-2457.811
2062.349									
0.930	0.3333	6762.750	961.059	8217.207	-1690.687	-2928.357	240.265	-416.151	2054.302
3558.155									
0.230	0.6667	3710.316	3956.768	6981.548	1743.278	634.501	343.543	-1948.328	-3280.254
-1193.916									

9.8.2 Response Function

The response function detail report contains axial load and bending moment RMS stresses and zero crossing period.

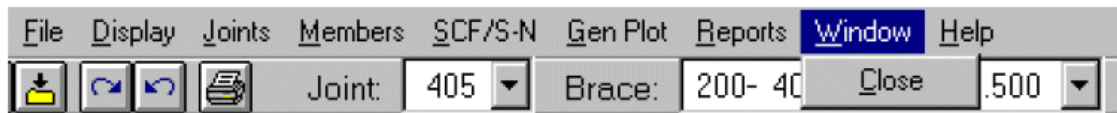
Response Function Detail Report						
Member 101- 203 Fatigue Load Case 1 Spectrum 1						
***** Axial Load *****		** Out-Plane Moment **		*** In-Plane Moment ***		
RMS	Zero Cross	RMS	Zero Cross	RMS	Zero Cross	
(KG)	(sec)	(CM-KG)	(sec)	(CM-KG)	(sec)	
-----	-----	-----	-----	-----	-----	-----
466.213	1.714	400.529	451.840	1.695	1.665	
		MOMENT		MOMENT		
FREQUENCY		AXIAL LOAD		OUT-PLANE	IN-PLANE	
HZ		KG **2/HZ		CM-KG **2/HZ	CM-KG **2/HZ	
-----		-----		-----	-----	-----
0.3611		65.110		2.728	76.051	
0.3750		440.079		23.890	457.900	
0.3889		2118.480		142.698	1965.001	
0.4028		7768.175		628.675	6426.693	
0.4167		22861.861		2168.366	16875.625	
0.4306		56261.289		6131.685	37060.078	
0.4444		119589.320		14739.152	70295.234	

9.9 JOINT CLASSES

Joint Classification Report for Joint 101				
LOAD	***** JOINT TYPE *****			
CASE	TY	X	KT	K1
1	0.891	0.000	0.000	0.109
2	0.895	0.000	0.000	0.105
3	0.906	0.000	0.000	0.094
4	0.568	0.000	0.000	0.432
5	0.864	0.000	0.000	0.136
6	0.879	0.000	0.000	0.121
7	0.882	0.000	0.000	0.118
8	0.884	0.000	0.000	0.116

The Joint Class report shows the fraction of the joint types applicable when determining SCFs for each load case used to develop the transfer function.

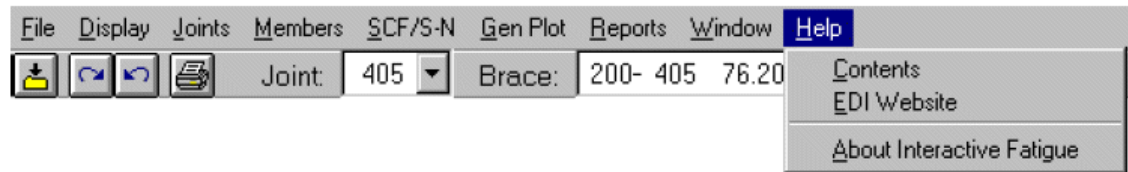
10 WINDOW MENU FEATURES



10.1 CLOSE

Close closes the current window.

11 HELP MENU FEATURES



11.1 CONTENTS

Contents brings up the Interactive Fatigue help file.

11.2 EDI WEBSITE

Goes to the Engineering Dynamics Inc. website (requires Internet connection).

11.3 ABOUT INTERACTIVE FATIGUE

About shows details of the version information for Interactive Fatigue and the copyright information.

12 Damage accumulation MENU FEATURES

This menu is only applicable when using a fatigue extract summation file. This file is auto generated by adding the extract line to the partial fatigue input file used in sacs wind turbine and GHBladed processes. Refer to Wind Turbine manual section 2 and 3 for details. User can create this file manually to sum up damages from different fatigue runs. It should be noted that in order to use this option all joint should be extracted by entering 'AL' in columns 14-15 of EXTRACT HEAD line.

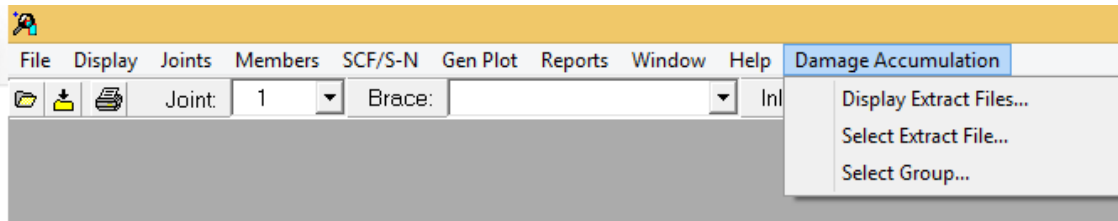
The Fatigue Extract summation files as shown below contains the following information:

SUMFTGIFT: Flag indicating a summation extract file
NCASES: Defines the number of simulation for analysis cases.
NGRPS: Defines the number of groups for analysis cases.
DIRECTORY: Directory location for fatigue extract file for a fatigue simulation
EXTRCT: Name of the extract file generated from above run
GROUP: Defines the group above run belongs to
Repeat the above (Directory to Group) for other simulations (total equals NCASES)
JOINTS: Defines the number of joints to be viewed in interactive fatigue (since not all joints need to be viewed)
Repeat all joint ID's desired (total equals JOINTS).

The example below shows a typical fatigue extract summation file for three simulations with 497 joints chosen to view.

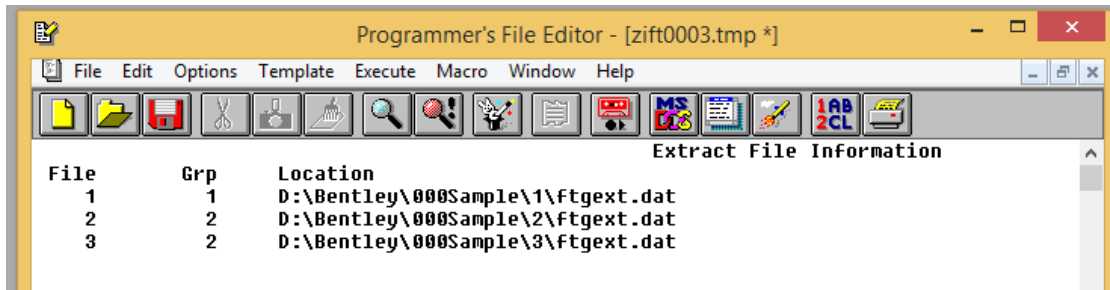
```
SUMFTGIFT
NCASES,3
NGRPS,2
DIRECTORY,"D:\Bentley\000Sample\1"
EXTRCT,"ftgext.dat"
GROUP,1
DIRECTORY,"D:\Bentley\000Sample\2"
EXTRCT,"ftgext.dat"
GROUP,2
DIRECTORY,"D:\Bentley\000Sample\3"
EXTRCT,"ftgext.dat"
GROUP,2
JOINTS,      497
  1
  3
  5
  7
 24
 71
 73
 75
 77
```

The following menu option is specifically used for such fatigue extract summation file:



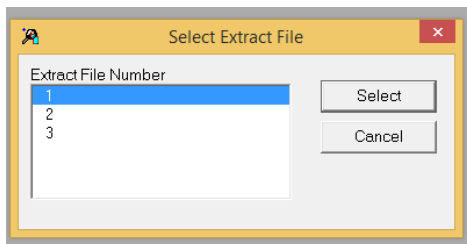
12.1 DISPLAY EXTRACT FILES

Contents brings up the list of extract file associated with this summation run.



12.2 Select Extract File

Brings up a dialog box to select which extract file to use to view results. By default results for the 1st file listed in the fatigue extract summation file are displayed.



12.3 Select group

Brings up a dialog box to select which files should be summed up when displaying results. The options are different groups used in fatigue extract summation file or ALL.

